

LECTURE 3.3



Joint Tests and Model Choice

ECONXXXXX Econometrics I

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PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

Notes

Weeks already covered are greyed; today's is marked.

Last Week's Block of Region Dummies

Region went into the earnings equation last week as a set of dummies, one region left out as the base.

Notes

Two minutes with your neighbour. Which coefficient gives the gap between two regions that are both in the model? And if none of the four dummies has a t-statistic above two, does region belong in the equation at all?

Lecture Plan

- **Testing Restrictions.** Asking about several coefficients at once, and why that is not the same as asking about each.
- **Choosing a Specification.** What adjusted R^2 charges for a variable, and what searching across specifications costs.
- **Multicollinearity.** What correlated regressors do to a standard error, and what they leave completely alone.

Notes

Reading: [chapter], and Leamer (1983), which is eleven pages, forty years old and still the sharpest statement of the problem in the middle section.

SECTION 1

Testing Restrictions

1. Testing Restrictions
2. Choosing a Specification
3. Multicollinearity

Why Separate T-Tests Cannot Answer a Joint Question

The goal of this section: ask a question about several coefficients at once, and see why the answer is not the separate answers collected together.

Two situations, and the t-statistics you already have settle neither:

1. Region belongs in the equation as a **block** or not at all. Four dummies give you four answers and no verdict.
2. Two coefficients, each unimpressive on its own, where dropping either one makes the other significant.

Four tests at the five per cent level is not a five per cent test of the four together. Each test gets its own chance to reject, and those chances accumulate. Hold on to that sentence: the middle section of today is the same arithmetic, run deliberately.

Restricted and Unrestricted Fits of the Same Model

A joint hypothesis is a claim that several coefficients are zero at the same time. Impose it and you are simply fitting a smaller equation. So fit both.

- The **unrestricted** fit: everything in.
- The **restricted** fit: the coefficients under test set to zero, everything else re-estimated around them.

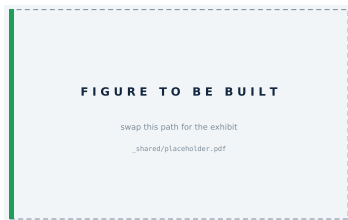


Figure 1: The same 120 people fitted twice. The restricted fit leaves more unexplained, and the difference between the two residual sums of squares is what the restriction cost.

Defining the F-Statistic From Two Residual Sums of Squares

What the next equation answers: how much fit did the restriction cost, per restriction, measured against how much is left unexplained anyway?

$$F = \frac{(RSS_r - RSS_u) / q}{RSS_u / (n - k - 1)} \quad (1)$$

- RSS_r : residual sum of squares from the restricted fit
- RSS_u : residual sum of squares from the unrestricted fit
- q : how many coefficients the restriction sets to zero
- n : the number of people in the sample
- k : the number of slopes in the unrestricted fit

RSS_r can never be smaller than RSS_u , for exactly the reason R^2 could not fall in 2.2. So the numerator is never negative, and the only question is whether the fit you gave up is more than chance would have

Joint Confidence Regions Versus Separate Intervals

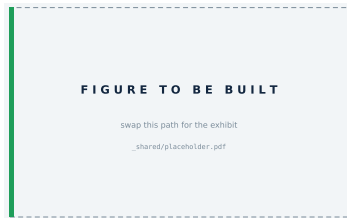


Figure 2: Two coefficients at once. Each interval on its own axis contains zero. The region that treats them jointly does not.

Nothing here is contradictory. The separate intervals answer separate questions, one coefficient at a time. The joint region answers the question you actually asked, which was about both.

Vote: Does Region Belong in the Equation?

The four region dummies from last week, each with a t-statistic of about 1.4. Does region belong in the equation?

Notes

Vote, argue with your neighbour, vote again. Then: what evidence would settle it?

Pause

Notes

Two minutes. Write down a hypothesis about this earnings equation that a t-test cannot express. Then compare with your neighbour.

SECTION 2

Choosing a Specification

1. Testing Restrictions
2. Choosing a Specification
3. Multicollinearity

Why R-Squared Cannot Choose Between Specifications

The goal of this section: choose between specifications, and be honest about what the choosing costs.

In 2.2 you voted on whether adding height in centimetres could lower R^2 . It cannot. Which means a rule that picks the specification with the highest R^2 picks the largest specification, every time, on every dataset, forever.

Notes

That is not a subtle flaw. It is the selection rule refusing to select.

Adjusted R-Squared and Its Degrees-of-Freedom Penalty

What the next equation answers: how well does this specification fit, per degree of freedom it spent getting there?

$$\bar{R}^2 = 1 - \frac{\text{RSS}/(n - k - 1)}{\text{TSS}/(n - 1)} \quad (2)$$

- RSS: residual sum of squares
- TSS: total sum of squares
- n : the number of people in the sample
- k : the number of slopes in this specification

What the penalty is worth

It can fall, which is the point of it, and it can go negative, which is occasionally instructive. What it cannot do is decide for you. Adding one variable raises $\bar{R}^2$ whenever that variable's t-statistic exceeds one in absolute value, and one is not a standard of evidence anybody would defend out loud.

The Smallest P-Value Across Twenty Specifications

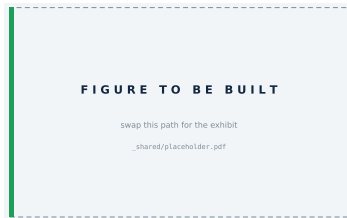


Figure 3: Twenty specifications fitted to an outcome with no signal in it, the smallest p-value kept, and the whole exercise repeated a thousand times. The dashed line is 0.05.

Notes

Freedman (1983) made the point with pure noise: screen enough regressors on data with nothing in it and the surviving equation looks entirely presentable. The arithmetic is correct throughout. What is wrong is the sentence attached to it.

Why P-Values Do Not Account for Specification Search

A p-value of 0.04 says something about a test chosen before the data were seen. If you fitted twenty specifications and reported the one that came in under five per cent, the number on your slide is not that p-value, and there is no correction printed anywhere in the output that repairs it.

Notes

Economics has measured this. Brodeur et al. (2016) and Brodeur et al. (2020) find published test statistics bunching just past the conventional thresholds, in quantities no honest distribution would put there. Simmons et al. (2011) show how few choices it takes; Gelman & Loken (2014) makes the harder version, that the search can happen without you ever feeling that you searched.

Research Practices That Make the Search Visible

- **Say what you will do before you look.** A pre-analysis plan fixes the specification and the outcome in advance (Olken, 2015).
- **Report the specifications you ran, not the one that worked.** Show the whole set and let the reader see the spread (Simonsohn et al., 2020).
- **Publish the code and the data,** so the search is visible to somebody who did not run it (Christensen & Miguel, 2018).

Not a rule against looking

Pre-specification costs something real, and nobody serious pretends otherwise – it forecloses things you would have learned by looking (Coffman & Niederle, 2015). Exploring is not dishonest. Exploring and then reporting it as confirmation is. A results table with exactly one column is a claim that exactly one specification was ever fitted, and you are entitled to ask whether that is true.

Vote: Would You Keep This Variable?

You add a variable. Schooling's coefficient barely moves, $\bar{R}^2$ rises by 0.004, and the new variable's t-statistic is 1.2.

Notes

Vote, argue with your neighbour, vote again. Then: does your answer change depending on whether you decided to add it before or after seeing the output?

Pause

Notes

Two minutes. Write down one specification decision you made in last week's lab that you could defensibly have made the other way. Then compare with your neighbour.

SECTION 3

Multicollinearity

1. Testing Restrictions
2. Choosing a Specification
3. Multicollinearity

Two Regressors Carrying Nearly the Same Information

The goal of this section: say precisely what correlated regressors do to an estimate, and then resist the reaction almost everybody has.

Schooling and years of potential experience move together in this population. Put both in and least squares has to work out which of them is moving earnings, using only the part of each that the other cannot account for – the partialling out from 3.1. When that leftover part is small, there is very little left to learn from.

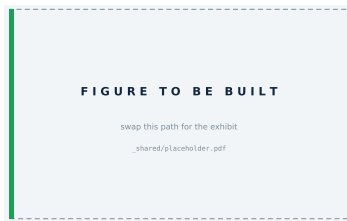


Figure 4: The correlation between two regressors raised from nought to 0.98. The standard error on the coefficient of interest climbs; the fitted line and R^2 barely move.

Collinearity Widens Standard Errors Without Breaking Assumptions

Collinearity breaks no assumption. The estimates are still unbiased, the fitted values and R^2 are hardly touched, Gauss-Markov still holds, and the standard errors are still right. They are simply large – because the data genuinely cannot separate two things that move together.

Notes

Goldberger (1991) made the point by inventing a matching disease, *micronumerosity*, meaning not enough observations, and giving it the same solemn treatment textbooks give collinearity. Both are statements that your sample carries less information than you wanted. Nobody writes a diagnostics section about having a small sample.

Why Dropping a Collinear Regressor Introduces Bias

The instinct is to drop one of the two. That does narrow the standard error, and it narrows it by declining to answer the question: the surviving coefficient now carries the omitted variable bias of 3.1, and it is no longer the partial effect you set out to estimate. You have traded a wide interval for a narrow one in the wrong place.

Notes

Lindner et al. (2020) trace how routinely this is done in published work, and what it does to the estimates. Vanhove (2021) works through the rest of the standard remedies and finds each of them quietly answering a different question from the one that was asked.

Variance Inflation Factors and Their Conventional Thresholds

- A variance inflation factor says how much wider a standard error is than it would be if that regressor were uncorrelated with the others. A real quantity, and worth looking at.
- The thresholds – five, ten – are conventions, not results. Whether an inflated standard error is a problem depends on whether the interval is still narrow enough to answer your question (O'Brien, 2007).
- What actually helps is what helped in 2.3: more people, or more spread in the regressor you care about.

Collinearity is a fact about your data, not a fault in your model. The regression is not misbehaving. It is reporting what it does not know.

Vote: Is This a Collinearity Problem?

Schooling and experience correlate at 0.85. Schooling's coefficient is 1.4 with a standard error of 0.9, and R^2 is 0.31.

Notes

Vote, argue with your neighbour, vote again. Then: what would you have to change about the data, rather than about the model, to narrow that interval?

What Would Have To Be True?

The standing question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

[Candidate this week: this is the only specification we fitted, and we chose it before we saw the estimate. Every claim the p-value makes rests on that, and nothing in the table can tell you whether it is so.]

Application

Search on purpose, on data you know has nothing in it.

Notes

Vote first: out of twenty specifications fitted to an outcome with no signal, how many come in under five per cent? Take the vote before anything is run.

Lab this week: generate an outcome unrelated to everything, fit twenty specifications, keep the smallest p-value, repeat. Then take the model that survived and fit it on 2.2's hold-out sample, which it has never seen.

Summary: Joint Tests, Model Choice and Collinearity

Today was three separate ideas, each doing one job:

1 Testing Restrictions.

An F-test compares a restricted fit against an unrestricted one, and answers a question the separate t-tests cannot.

2 Choosing a Specification.

Adjusted R-squared charges for a variable, but no fit statistic charges for the search that found the specification.

3 Multicollinearity.

Correlated regressors widen standard errors and break nothing, and dropping one buys its precision with bias.

Notes

Next week: the Gauss-Markov assumptions from 2.3, and how you would notice one of them failing in data you did not simulate yourself.

Before Lecture 4.1

- **Read** [chapter], and Leamer (1983), which is short.
- **Do** the appendix derivations. There are two, and the first is where the F-statistic stops being a formula and becomes a comparison.
- **Try** the specification-search app: set the true effect to zero, fit twenty specifications, and watch the smallest p-value.

Problem set 3

[Out this week, due week n. Covers 3.2 and 3.3, and carries one question back from problem set 2.]

Thank you

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BACKUP

Appendix

Derivation: The F-Statistic as Restricted Against Unrestricted Fit

Fit the equation twice, once with the restriction imposed and once without, and write the cost of the restriction as a ratio of two separate estimates of the same error variance.

Notes

Four steps. Write the restricted equation as the unrestricted one with q coefficients set to zero; show that the restricted residual sum of squares cannot be the smaller of the two; divide the difference by q and the unrestricted residual sum of squares by $n - k - 1$; show that under the null both of those are estimating the same error variance, so their ratio follows an F distribution with q and $n - k - 1$ degrees of freedom. The step that does the work is the last one, and it is the step that needs the assumptions of 2.3 and normal errors on top of them.

[Full derivation. Set as homework; worked solution published with the set.]

Derivation: When Adjusted R-Squared Rises

Show that adding one variable raises $\bar{R}^2$ if and only if that variable's t-statistic is greater than one in absolute value.

Notes

Two steps and no new machinery. Write $\bar{R}^2$ for the small and the large equation in terms of their residual sums of squares, take the difference, and recognise what is left as the square of a t-statistic compared with one. The result is the justification for the caveat in section two: the threshold $\bar{R}^2$ imposes is real, and it is far below anything you would call evidence.

[Full derivation. Set as homework; worked solution published with the set.]

Notes and Tools

- `anova(fit_restricted, fit_unrestricted)` in R, which is the whole of section one in one line.
- `car::linearHypothesis()` for restrictions that are not “all zero”.
- `summary(fit)$adj.r.squared`, and `car::vif()` – read the second alongside O’Brien (2007) rather than against a threshold.
- The specification-search script, and 2.2’s hold-out sample.
- On where this module sits in the wider argument about what a first econometrics course should spend its hours on: Angrist & Pischke (2017).

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